

SOZIALZYNC

Developer & SuperAdmin Pending Completion Guide

Everything required to go from dev build → production-ready SaaS

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AI-Powered YouTube Content Operating System

■ Overview & Priority Order

This guide lists every pending configuration, environment variable, OAuth integration, database migration, and feature wire-up required before Sozialzync is fully production-ready. Items are ordered from most-blocking to least-blocking.

#	Task	Notes	Status
1	Set up PostgreSQL database & run migrations	Local Docker or hosted Postgres — all Prisma models need to be applied	Pending
2	Configure all required env vars	See Section 2 — 25+ variables across API and Web apps	Pending
3	YouTube OAuth 2.0 setup	Google Cloud Console — scopes: youtube.upload, youtube.readonly	Pending
4	ElevenLabs API key	Free tier: 10k chars/month. Used for voice synthesis & character voices	Pending
5	OpenAI API key	Required for TTS, DALL-E 3 thumbnails, AI script generation	Pending
6	Jamendo client_id	Free account at developer.jamendo.com — royalty-free music discovery	Pending
7	Pixabay API key	Free at pixabay.com/api — royalty-free images and music	Pending
8	Pexels API key	Free at pexels.com/api — royalty-free photo search	Pending
9	Unsplash Access Key	Free at unsplash.com/developers — royalty-free photo search	Pending
10	Clerk Auth integration	Add Clerk middleware and provider to replace custom JWT flow	Optional
11	Sentry DSN configuration	Error tracking & on-call alerts. Add to both web and API envs	Pending
12	Vercel deployment env vars	Mirror all API env vars in the Vercel dashboard for the web app	Pending
13	Create first SuperAdmin user	Run seed script or update user.role = 'SUPER_ADMIN' in DB	Pending
14	Configure AI provider fallbacks	Set primary + fallback providers in SuperAdmin → AI Providers page	Pending
15	Set up Cloudflare R2 or local storage	STORAGE_BACKEND=r2 needs R2 bucket + credentials	Optional

16	BullMQ / Redis setup	Required for background AI jobs — use Upstash Redis free tier	Pending
17	SMTP / email provider	For OTP emails, password reset, notifications	Pending
18	Stripe payment setup	For billing and wallet top-ups — webhook endpoint required	Optional
19	Run full E2E test suite	pnpm e2e — all navigation, orgs, copilot, publishing tests	Pending
20	Lighthouse CI score ≥ 80	Accessibility and performance gates in .lighthouserc.yml	Pending
21	OWASP ZAP security scan	Passive + active scan in CI — review and triage findings	Pending

1 Database Setup

1.1 Local Development (Docker)

Run a local PostgreSQL instance with Docker Compose:

```
docker run -d --name cf-postgres \  
-e POSTGRES_USER=cfuser \  
-e POSTGRES_PASSWORD=cfpass \  
-e POSTGRES_DB=creatorforce \  
-p 5432:5432 postgres:16-alpine
```

Then set DATABASE_URL in apps/api/.env:

```
DATABASE_URL=postgresql://cfuser:cfpass@localhost:5432/creatorforce
```

1.2 Apply All Migrations

From the apps/api directory, run:

```
cd apps/api  
npx prisma migrate deploy  
npx prisma generate
```

1.3 Migrations to Apply (in order)

- 20260701000000_init — Base schema (User, Project, Channel, Video, etc.)
- 20260710000000_copilot — Copilot conversation model
- 20260722000000_optional_channel_platform — Channel platform field nullable
- 20260729000001_user_provider_configs — AI provider config per user
- 20260729000002_music_tracks — MusicTrack model for royalty-free library
- 20260730000001_publishing_status — PublishingStatus enum + publishedAt
- 20260805000001_add_character_model — Character Studio model (NEW)

1.4 Seed First SuperAdmin

After migrations, create the first admin user. Either register via the UI at /register and then update the role in the DB:

```
-- Run in psql after registering:  
UPDATE "User" SET role = 'SUPER_ADMIN'  
WHERE email = 'your-admin@email.com';
```

2 Environment Variables

Set these in apps/api/.env (backend) and apps/web/.env.local (frontend). Never commit .env files. In production, use Vercel environment variables dashboard or a secret manager.

Core Backend (apps/api/.env)

Environment Variable	Required	Source	Description
DATABASE_URL	✓	Your Postgres provider	PostgreSQL connection string
JWT_SECRET	✓	Generate: openssl rand -hex 32	Signs auth tokens — keep secret
REFRESH_SECRET	✓	Generate: openssl rand -hex 32	Signs refresh tokens
REDIS_URL	✓	Upstash Redis / local Redis	Required for BullMQ job queues

AI Providers

Environment Variable	Required	Source	Description
OPENAI_API_KEY	✓	platform.openai.com	TTS, DALL-E 3 thumbnails, GPT-4o scripts
ANTHROPIC_API_KEY	✓	console.anthropic.com	Claude — primary AI for scripts, compliance
GEMINI_API_KEY	—	aistudio.google.com	Fallback AI provider (optional)
ELEVENLABS_API_KEY	✓	elevenlabs.io	Voice synthesis — character voices, TTS

YouTube OAuth (Google Cloud Console)

Environment Variable	Required	Source	Description
YOUTUBE_CLIENT_ID	✓	Google Cloud Console	YouTube Data API v3 OAuth client
YOUTUBE_CLIENT_SECRET	✓	Google Cloud Console	YouTube OAuth secret
YOUTUBE_REDIRECT_URI	✓	e.g. http://localhost:4007/...	Must match Google Console exactly

Royalty-Free Media APIs (all free tier)

Environment Variable	Required	Source	Description
JAMENDO_CLIENT_ID	✓	developer.jamendo.com (free)	Royalty-free music discovery
PIXABAY_API_KEY	✓	pixabay.com/api (free)	Royalty-free images + music
PEXELS_API_KEY	✓	pexels.com/api (free)	Royalty-free photo search

UNSPLASH_ACCESS_KEY	✓	unsplash.com/developers (free)	Royalty-free photo search
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Email / SMTP

Environment Variable	Required	Source	Description
SMTP_HOST	✓	e.g. smtp.sendgrid.net	Email delivery host
SMTP_PORT	✓	Usually 587	Email port (TLS)
SMTP_USER	✓	From your email provider	SMTP auth username
SMTP_PASS	✓	From your email provider	SMTP auth password
EMAIL_FROM	✓	noreply@yourdomain.com	Sender address for system emails

File Storage (Cloudflare R2 — optional)

Environment Variable	Required	Source	Description
STORAGE_BACKEND	—	local or r2	Default: local file storage
R2_ACCOUNT_ID	—	Cloudflare dashboard	Required if STORAGE_BACKEND=r2
R2_ACCESS_KEY_ID	—	Cloudflare dashboard	R2 credentials
R2_SECRET_ACCESS_KEY	—	Cloudflare dashboard	R2 credentials
R2_BUCKET_NAME	—	Your R2 bucket name	Where media files are stored

Payments (Stripe — optional)

Environment Variable	Required	Source	Description
STRIPE_SECRET_KEY	—	dashboard.stripe.com	Payment processing — wallet top-ups
STRIPE_WEBHOOK_SECRET	—	Stripe CLI / dashboard	Validates webhook events

Frontend (apps/web/.env.local)

Environment Variable	Required	Source	Description
NEXT_PUBLIC_API_URL	✓	e.g. http://localhost:4007	Backend API base URL
NEXT_PUBLIC_APP_URL	✓	e.g. http://localhost:3007	Frontend base URL
NEXT_PUBLIC_SENTRY_DSN	—	sentry.io project DSN	Frontend error tracking
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY	—	clerk.com dashboard	Clerk auth (optional overlay)

Backend Monitoring & Auth

Environment Variable	Required	Source	Description
SENTRY_DSN	—	sentry.io project DSN	Backend error tracking
SENTRY_AUTH_TOKEN	—	sentry.io user token	For release tracking in CI
CLERK_SECRET_KEY	—	clerk.com dashboard	Clerk backend validation

3 YouTube OAuth 2.0 Setup

Sozialzync publishes videos to YouTube on behalf of creators. This requires a Google Cloud project with the YouTube Data API v3 enabled.

3.1 Google Cloud Console Steps

- 1 Go to console.cloud.google.com → New Project → Name: "Sozialzync"
- 2 APIs & Services → Enable APIs → Search "YouTube Data API v3" → Enable
- 3 APIs & Services → Credentials → Create Credentials → OAuth 2.0 Client ID
- 4 Application type: Web application
- 5 Authorised redirect URIs: add both:
 - <http://localhost:4007/api/v1/auth/youtube/callback> (development)
 - <https://api.yourdomain.com/api/v1/auth/youtube/callback> (production)
- 6 Copy Client ID → YOUTUBE_CLIENT_ID env var
- 7 Copy Client Secret → YOUTUBE_CLIENT_SECRET env var
- 8 OAuth consent screen → Add scopes:
 - [.../auth/youtube.upload](#)
 - [.../auth/youtube.readonly](#)
 - [.../auth/youtube.force-ssl](#)
- 9 For production: submit OAuth app for Google verification (required to upload as non-test users)

3.2 Test the Flow

Once env vars are set, the OAuth flow is at:

GET </api/v1/auth/youtube> → redirects to Google

GET </api/v1/auth/youtube/callback> → stores tokens + redirects to /publish

Tokens are stored in the ChannelAccount table per user. The publishing flow automatically refreshes expired tokens.

4 Royalty-Free Media API Registration

4.1 Jamendo (Free Music — 600k+ Creative Commons tracks)

- 1 Go to developer.jamendo.com
- 2 Click "Register" → create a free account
- 3 Go to "API Keys" → "Create new application"
- 4 Copy the `client_id` → set as `JAMENDO_CLIENT_ID`
- 5 Rate limit: 3,000 requests/day (free tier)

4.2 Pixabay (Free Images + Music)

- 1 Go to pixabay.com → sign up for free
- 2 Go to pixabay.com/api/ → copy your API key
- 3 Set as `PIXABAY_API_KEY`
- 4 Rate limit: 5,000 requests/hour (free tier)

4.3 Pexels (Free Stock Photos)

- 1 Go to pexels.com/api/ → sign up
- 2 Click "Your API Key" → copy key
- 3 Set as `PEXELS_API_KEY`
- 4 Rate limit: 200 requests/hour, 20,000/month (free tier)

4.4 Unsplash (Free High-Quality Photos)

- 1 Go to unsplash.com/developers → "New Application"
- 2 Fill in application details → accept terms
- 3 Copy "Access Key" → set as `UNSPLASH_ACCESS_KEY`
- 4 Rate limit: 50 requests/hour (demo), unlimited after approval

4.5 ElevenLabs (Voice Synthesis — Character Voices)

- 1 Go to elevenlabs.io → Sign up free
- 2 Profile → API Key → copy key
- 3 Set as `ELEVENLABS_API_KEY`
- 4 Free tier: 10,000 characters/month
- 5 Character presets use OpenAI TTS by default — ElevenLabs is optional but improves quality

5 Redis Setup (Required for Background Jobs)

All AI generation tasks (scripts, thumbnails, video generation, TTS) run as BullMQ background jobs. Redis is required.

5.1 Option A: Upstash Redis (Free Hosted)

- 1 Go to console.upstash.com → Create Database
- 2 Choose region closest to your API server
- 3 Copy the Redis URL (format: `redis://:password@host:port`)
- 4 Set as `REDIS_URL` in `apps/api/.env`
- 5 Free tier: 10,000 commands/day

5.2 Option B: Local Redis (Docker)

```
docker run -d --name cf-redis -p 6379:6379 redis:7-alpine
REDIS_URL=redis://localhost:6379
```

6 New Features Built — Character Studio

The Character Studio was implemented in this session. Here is what was built and what still needs to be wired up post-deploy.

6.1 What Was Built

- **Character voices with FFmpeg audio effects** — 8 effect presets: natural, cartoon, chipmunk, villain, giant, robot, whisper, echo. Each applies FFmpeg filter chains to pitch-shift and process TTS output.
- **8 built-in character presets** — Cartoon Hero, Evil Villain, Funny Robot, Wise Elder, Anime Star, Mysterious Voice, Chipmunk Charlie, Pixel Gamer. Each has a personality description, voice settings, and video style.
- **DiceBear avatar integration** — 12 avatar art styles (Avataaars, Bottts, Fun Emoji, Pixel Art, etc.) served free from api.dicebear.com — no API key needed.
- **Character Studio page** at `/studio/characters` — 3 tabs: My Characters (CRUD), Presets (browse & add), Create Custom (full form).
- **Voice preview streaming** — POST `/api/v1/characters/preview-voice` returns MP3 with voice effect applied, playable inline.
- **6 video styles** — Realistic, Cartoon, Animation, Anime, Movie, Pixel Art — stored per character and used as generation prompts for future video AI.
- **Image & Assets Studio** at `/studio/assets` — search Pexels, Unsplash, Pixabay, Openverse in parallel. Download and use in projects.
- **AI Thumbnail Generator** at `/studio/assets` (AI Thumbnail tab) — generates DALL-E 3 thumbnail images at YouTube's 16:9 ratio with style variations.

6.2 Post-Deploy Wiring Required

- Run migration `20260805000001_add_character_model` against production DB
- Ensure `OPENAI_API_KEY` is set — required for voice preview (OpenAI TTS) and thumbnail generation (DALL-E 3)
- `ELEVENLABS_API_KEY` optional — for higher quality voices on ElevenLabs provider
- FFmpeg must be installed on the API server: `sudo apt-get install ffmpeg` (for voice effects)
- Add `PEXELS_API_KEY`, `UNSPLASH_ACCESS_KEY`, `PIXABAY_API_KEY` for image search in assets studio

7 Music & Voice Libraries

Built in prior sessions. Requires env vars to activate.

- **Music Library** at /studio/music — Browse tab searches Jamendo + Pixabay for royalty-free tracks. Requires JAMENDO_CLIENT_ID and PIXABAY_API_KEY.
- **AI Music Match** — analyzes project script and auto-selects best matching track via Claude AI. Requires ANTHROPIC_API_KEY.
- **Voice Library** at /studio/voices — fetches ElevenLabs premade voices + 6 OpenAI voices. Preview and import to projects.
- **AI Voice Select** — auto-picks voice based on script mood and content. Requires ANTHROPIC_API_KEY + ELEVENLABS_API_KEY.

8 SuperAdmin Configuration

After setting the first user to SUPER_ADMIN role, log in and complete these steps in the admin panel at /admin.

8.1 Admin → AI Providers Tab

- Set Primary LLM Provider: Anthropic (Claude claude-sonnet-4-6 recommended)
- Set Fallback Provider: OpenAI (GPT-4o) — automatically used on Anthropic 429/5xx
- Configure per-provider rate limits and token budgets
- Enable/disable Gemini as secondary fallback

8.2 Admin → Platform Settings

- Set platform name, logo, support email
- Configure max video generation concurrency (default: 3 concurrent per user)
- Set compliance check strictness level (strict/standard/relaxed)
- Enable/disable beta features: Character Studio, AI Thumbnail, Auto-Publish

8.3 Admin → Users Tab

- Verify first user accounts manually if email verification is not set up
- Set role (MEMBER / ADMIN / SUPER_ADMIN) per user
- Monitor AI usage per user — visible in Admin → AI Usage tab

9 Deployment Checklist

9.1 Frontend — Vercel

- 1 Frontend is already deployed to Vercel (project: creatorforce-ai)
- 2 Go to vercel.com → Project → Settings → Environment Variables
- 3 Add ALL variables from Section 2 (Frontend + Backend monitoring)
- 4 NEXT_PUBLIC_API_URL must point to your production API domain
- 5 Trigger redeploy after adding env vars

9.2 Backend API — Docker / Railway / Render

- 1 Build: `docker build -f infra/Dockerfile.api -t sozialzync-api .`
- 2 Required: PostgreSQL, Redis URLs in environment
- 3 Required: FFmpeg in the Docker image (add to Dockerfile: `RUN apt-get install -y ffmpeg`)
- 4 Port: 4007 (configured in NestJS main.ts)
- 5 Run migrations on deploy: `npx prisma migrate deploy`
- 6 Health check endpoint: `GET /health`

9.3 CI/CD Secrets (GitHub Actions)

- CODECOV_TOKEN — from codecov.io (test coverage reports)
- SENTRY_AUTH_TOKEN — for release tracking in CI
- All standard env vars — set in GitHub → Settings → Secrets → Actions

10 Quick Start Commands

Run these from the repository root after setting env vars:

Install dependencies

```
pnpm install
```

Start all services (API + Web)

```
pnpm dev
```

API runs on :4007, Web on :3007

Run database migrations

```
cd apps/api && npx prisma migrate deploy && npx prisma generate
```

Run E2E tests

```
pnpm e2e
```

Run unit tests with coverage

```
pnpm test -- --coverage
```

Build for production

```
pnpm build
```

Type-check all packages

```
pnpm typecheck
```

Lint

```
pnpm lint
```

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